

Zainab Soomro

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EDUCATION

University of Southern California, Los Angeles, CA

Jan 2022 - Dec 2023

Master of Science in **Applied Data Science**

Received **Fulbright Scholarship**

Coursework: Machine Learning, Data Mining, Database Management, Medical Imaging Informatics

Lahore University of Management Sciences, Lahore, Pakistan

Aug 2016 - May 2020

Bachelor of Science (Honors) in **Management Science**, Minor: **Computer Science**

Dean's Honor List for Academic Year 2016-2017, Fall 2017 & Spring 2020

Coursework: Computer Vision, Artificial Intelligence, Probability & Statistics, Linear Algebra, Business Analytics

WORK EXPERIENCE

Data Scientist | Indus Hospital and Health Network, Karachi, Pakistan

May 2025 – Present

- **Established** the hospital's Data Science department, collaborating with multi-disciplinary teams to streamline workflows
- **Structured 1M+ ED visits into NCD categories (>95% accuracy)**, unlocking new insights through survival analysis, trend monitoring, and real-time dashboards.
- Developed and evaluated **clustering** (k-Means, k-Prototypes, DBSCAN) on **7,000+ toxicology ED visits** and stratified **forecasting models** (ARIMA, SARIMAX, XGBoost, Prophet, TFT) on **1M+ ED records**.
- **Guided student research** on applied health AI, including a **bilingual diabetes chatbot** and a **crisis ambulance app**.

Researcher | iLab USC (Dr. Laurent Itti), Los Angeles, CA

Jan 2023 – Oct 2023

- Led **multimodal** research integrating non-invasive wearable EEG and free viewing eye-tracking data to aid in **early detection of ALS** leveraging machine learning models.
- Completed **phase one**: shortlisting and set-up of devices, sample collection of data, and exploratory data analysis.

Implementation and Activation Lead (IT & Digital) | Reckitt (Global), Karachi, Pakistan

Jan 2021 - Dec 2021

- Orchestrated **end-to-end implementation** of an advanced analytical chatbot solution, servicing over **100 users** in Pakistan and catering to a Reckitt Health client base of **70+ users**.
- Spearheaded a transformative shift from localized data collection to **central automated data aggregation** for chatbot, resulting in a faster expansion to **10 markets within one year**, a feat previously estimated to take six years.

PUBLICATIONS

Sentiment, Count and Cases: Analysis of Twitter discussions during COVID-19 Pandemic | BESC 2020

Jul 2020

- Performed **sentiment analysis** on **18 million tweets** related to **COVID-19** from Mar 1, 2020 to May 31, 2020.
- Discovered a significant **positive correlation** between the **number of tweets** mentioning Italy, USA, and UK and the **daily increase in COVID-19** cases in these countries.

Analyzing Brexit's impact using sentiment analysis and topic modeling on Twitter discussion | D.Go 2020

Jan 2020

- Nominated for **best paper award** in Social Media and Government Track.
- Analyzed **16 million tweets** about Brexit from Sept 11, 2019 to Oct 31, 2020 using **topic modelling** and **sentiment analysis**.
- Determined a **strong positive correlation** between the value of the Pound Sterling and public sentiment regarding Brexit on Twitter. Results also established that topics discussed on twitter are **highly representative of real-life** events.

PROJECTS

Recommendation System: Collaborative Filtering for Yelp

Mar 2023 - Apr 2023

- Designed **Item-based CF model** with **Pearson similarity** with **1.02 RMSE** to predict user-business ratings using **Hadoop, Spark, and Python**; **Reduced RMSE to 0.96** using **feature engineering and ensemble learning (CF and XGBoost Regression)**.
- Fine tuned recommendation system with **user-based average and ensemble model** to tackle **cold start and biased ratings**.

Big Data Distribution System: Hadoop Emulation (Awarded best project)

Aug 2022 - Dec 2022

- Engineered end-to-end resilient **distributed system in Python Flask**, employing **map-reduce** paradigm and for efficient storage, partitioning for **30% faster analysis of large datasets**.
- Designed and implemented a **scalable User Interface (UI) using Angular**, enabling **2x better user interactions and data analysis** while supporting both **SQL and NoSQL** data architectures.

SKILLS

Python, C++, R, Apache Spark | PyTorch, Tensorflow, Hadoop, NumPy, Pandas, Scikit-learn, OpenCV, Vader | SQL, MySQL, MongoDB, Firebase | GitHub, PowerBI, AWS | Regression, Classification, SVM, Naive Bayes, Decision Trees, REST APIs, NLP, Optimization